

Desfire EV1 Card API

Development Documentation

V1.0.0

Update Records

Date	Description
2024-07-04	First Edition Release

Contents

- 1 NFC Class
- 2 Password Authentication
- 3 Modify the key
- 4 Create an App
- 5 Create a File
- 6 Delete existing App
- 7 Delete an existing File
- 8 Format the card
- 9 Get the APPID of all existing APPs
- 10 Get the FileNo of all existing files
- 11 Select the corresponding APPLICATION by APPID
- 12 Read data from the target file
- 13 Write data to the target file

1 NFC Class

```
/**
 * @Function NFC Construction method.
 *
 * @Parameter
 * 1.context: Context
 *
 * @Return Nfc: Nfc object
 *
 * */

public Nfc(Context context)
```

2 Password Authentication

```
/**
 * @Function Password authentication. You need to select a directory before
calling this method.
 *
 * @Parameter
 * 1.KeyNo: The number of the Key. 1 byte, KeyNo = 0 means master key.
 * 2.keytype: The type of key. 1 byte, for expansion, currently needs to be set to
0.
 * 3.desKey: The key. 16 bytes, The key which encryption and decryption uses a 16-
byte 2K3DES algorithm. The default password for the new card is 16 bytes of 0x00.
 *
 * @Return 0, success; Non 0, failed
 *
 * */

public synchronized int md_auth(byte KeyNo, byte keytype, byte[] desKey) throws
CommonException
```

3 Modify the key

```
/**
 * @Function Modify the key. Password authentication is required before calling
this method.
 *
 * @Parameter
 * 1.KeyNo: The number of the target key. 1 byte.
 * 2.desKey: The new key.16 bytes.
```

```

*
* @Return 0, success; Non 0, failed
*
* */

public synchronized int md_change_key(byte KeyNo, byte[] desKey) throws
CommonException

```

4 Create an App

```

/**
 * @Function Create an App. You must first select the system directory before
calling this method.
 *
 * @Parameter
 * 1.appid: Each APPLICATION needs to be associated with an APPID. 4 bytes, only
the low 24bit is valid, that is, the maximum value is 0xFFFFFF.
 * 2.KeySetting: The permission settings of key.1 byte,and the default value is
0x0F.
 * 3.MaxKeyNo: The maximum number of Keys that the APPLICATION can contain.1 byte.
 *
 * @Return 0, success; Non 0, failed
 *
 * */

public synchronized int md_create_app(int appid, byte KeySetting, byte MaxKeyNo)
throws CommonException

```

5 Create a File

```

/**
 * @Function Create a File.You need to select an existing target directory Before
calling this method.
 *
 * @Parameter
 * 1.FileNo: Each file needs to be associated with a FileNo. 1 byte, the maximum
number of files that can be created in each APPLICATION is 16, so the value range
is 0x00-0x0F.
 * 2.commset: Communication encryption. 1 byte, currently only supports commset =
0 for MACed encrypted communication.
 * 3.inBuf: Indicates which keys the file needs to verify for read and write
permissions. 2 bytes, please refer to Desfire's specification for specific
meaning. If it is not clear, it can be set to 0x0000.
 * 4.inLen: The byte array length of inBuf.
 * 5.filesize: The size of the file created. 4 bytes, only low 24bit effective.
 *

```

```
* @Return 0, success; Non 0, failed
*
* */
```

```
public synchronized int md_create_std_data_file(byte FileNo, byte commset, byte[]
inBuf, int inLen, int filesize) throws CommonException
```

6 Delete existing App

```
/**
 * @Function Delete existing App.You need to select the system directory and pass
password authentication before calling this method.
 *
 * @Parameter
 * 1.appid: APPID corresponding to the target App.
 *
 * @Return 0, success; Non 0, failed
 *
 * */

public synchronized int md_del_app(int appid) throws CommonException
```

7 Delete an existing File

```
/**
 * @Function Delete an existing File. You need to select the target directory and
pass password authentication before calling this method.
 *
 * @Parameter
 * 1.FileNo: FileNo corresponding to the target file
 *
 * @Return 0, success; Non 0, failed
 *
 * */

public synchronized int md_del_file(byte FileNo) throws CommonException
```

8 Format the card

```
/**
 * @Function Format the card.You need to select the system directory and pass
password authentication before calling this method.
```

```

*
* @Return 0, success; Non 0, failed
*
* */

public synchronized int md_factory_reset() throws CommonException

```

9 Get the APPID of all existing APPs

```

/**
 * @Function Get the APPID of all existing APPLICATIONs.You must first select the
 * system directory before calling this method.
 *
 * @Return Byte array containing all APPIDs
 *
 * */

public synchronized byte[] md_get_app_ids() throws CommonException

```

10 Get the FileNo of all existing files

```

/**
 * @Function Get the FileNo of all existing files.You need to select an existing
 * target directory Before calling this method.
 *
 * @Return Byte array containing all FileNos
 *
 * */

public synchronized byte[] md_get_file_ids() throws CommonException

```

11 Select the corresponding APPLICATION by APPID

```

/**
 * @Function Select the corresponding APPLICATION by APPID. When performing
 * directory level related operations, such as creating a directory, you must first
 * select the directory with APPID = 0. This directory is the default system
 * directory and cannot be deleted or created.
 *
 * @Parameter
 * 1.appid: APPID corresponding to the target APPLICATION.
 *

```

```
* @Return 0, success; Non 0, failed
*
* */
```

```
public synchronized int md_select_app(int appid) throws CommonException
```

12 Read data from the target file

```
/**
 * @Function Read data from the target file.You need to select the target
directory and pass password authentication before calling this method.
 *
 * @Parameter
 * 1.FileNo: FileNo corresponding to the target file.
 * 2.offset: Read the offset of the position. 4 bytes, only low 24bit effective.
 * 3.inLen: Read the length of the data.4 bytes, only low 24bit effective.
 *
 * @Return The data content read. No more than 256 bytes
 *
 * */

public synchronized byte[] md_read_data(byte FileNo, int offset, int inLen) throws
CommonException
```

13 Write data to the target file

```
/**
 * @Function Write data to the target file.You need to select the target directory
and pass password authentication before calling this method.
 *
 * @Parameter
 * 1.FileNo: FileNo corresponding to the target file.
 * 2.offset: The offset of the write location.4 bytes, only low 24bit effective.
 * 3.length: The length of the data written.4 bytes, only low 24bit effective.
 * 4.inBuf: The data content written. No more than 256 bytes
 * 5.inLen: The byte array length of inBuf.
 *
 * @Return 0, success; Non 0, failed
 *
 * */

public synchronized int md_write_data(byte FileNo, int offset, int length, byte[]
inBuf, int inLen) throws CommonException
```